

"Rules or Data? Gotta Catch 'Em All!": A Tangible Game for Youth AI Literacy

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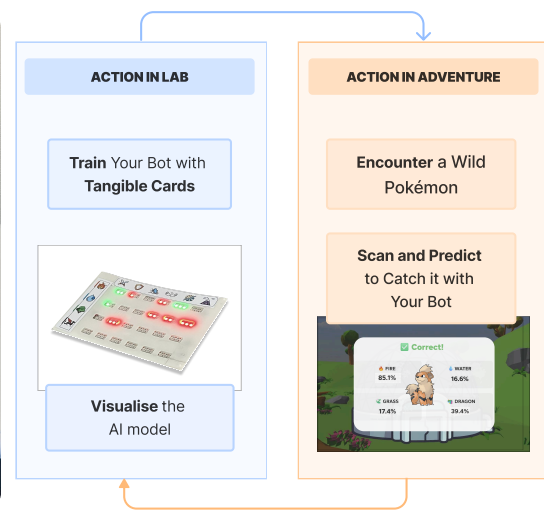


Figure 1: We introduce *Pokémon AI Adventure*, a tangible interaction game designed to support adolescents developing AI literacy. Learning how to play the game involves learning to both train small robots using physical cards and test them by capturing mythical creatures, called Pokémon. In the process, the game helps learners develop a conceptual connection between human design choices and AI behavior by contrasting gameplay with rule-based and data-driven systems.

Abstract

Adolescents often learn about AI through single-model demonstrations, such as training a classifier, which can blur how rule-based and data-driven systems differ and why those differences matter for errors, bias, and trust. We present *Pokémon AI Adventure*, a tangible, narrative-driven learning game in which students build and compare two robot partners: RuleBot, which follows feature-based rules, and DataBot, which learns from examples. To complete zone challenges and capture Pokémon, students assemble rule sets by selecting and ordering physical Clue Cards, and curate training datasets using Package Cards while managing noise, imbalance, and missing Pokémon types. Through these activities, the game

supports learning about how rules and data shape model behavior, how accuracy and bias arise from human choices, and how to develop more reasoned trust in different AI systems.

CCS Concepts

• **Human-centered computing** → User interface toolkits; • **Social and professional topics** → Computing education.

Keywords

AI literacy, child-computer interaction, tangible learning, educational games, rule-based systems, machine learning



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1 Introduction

Adolescents increasingly encounter AI systems in everyday settings: from educational tools to media platforms and automated recommendations [1]. This shift has intensified interest in AI literacy for young people, with recent research emphasizing that learners should understand not only what AI can do, but also how AI systems are designed, trained, and used responsibly [5, 31].

Most introductory AI learning experiences center on a single paradigm, typically a simplified machine learning pipeline such as training an image classifier from examples [4, 7, 10, 12, 16, 20, 33]. While popular tools and curricula make this workflow accessible and engaging, they can also promote a monolithic view of AI that obscures meaningful distinctions between approaches [35]. Prior work in child–AI interaction shows that youth often attribute unexpected AI behavior to “randomness” or the system “being weird,” rather than tracing errors back to concrete human choices in features, rules, or training data [6, 11].

A key missing piece is support for *contrastive understanding*: helping adolescents explicitly differentiate rule-based and data-driven approaches [9, 15]. Rule-based systems follow human-authored decision logic. In contrast, data-driven systems infer patterns from training examples. These two approaches lead to different failure modes and offer differing opportunities for model debugging and accountability [22, 23]. Yet, few learning experiences make this contrast concrete through iterative construction, testing, and revision across both paradigms.

Research in HCI and Child–Computer Interaction suggests that tangible and embodied interaction can help learners externalize their reasoning, inspect intermediate computational states, and revise their mental models through hands-on manipulation [8]. This perspective motivates learning activities that transform abstract AI concepts into inspectable objects and actions, rather than leaving them as invisible computations [13]. Complementary work in computing education further highlights how tangible and unplugged approaches, often combined with storytelling, support engagement and conceptual access through physically grounded representations [3, 14].

In this work, we introduce *Pokémon AI Adventure*: a tangible, narrative-driven interactive experience for adolescents. Participants take on the role of Trainers who train and compare two robots—*RuleBot* and *DataBot*—by physically manipulating them during gameplay. Using *Clue Cards*, participants create sets of explicit rules to control RuleBot’s predictions. For DataBot, *Package Cards* curate training datasets under constraints such as noise, imbalance, and missing categories. Through progressive disclosure, the game sequences challenges encountered in each zone. This experience invites hands-on comparison of rule-based and data-driven approaches to AI. In turn, progress through the game directly exposes differences in model behavior, failure modes, and accountability.

We present *Pokémon AI Adventure* as an interactive demo that showcases how tangible, contrastive interaction can support experiential understanding of AI systems. Preliminary playtesting with adolescents informed the design of the game mechanics and interaction flow, confirming the feasibility of the tangible setup, and the clarity of the contrast between AI paradigms. At CHI, the demo enables attendees to engage directly with the system, observe

how different design choices in rules and data shape AI behavior in real time, and reflect on the role of human decisions in model performance, bias, and trust.

2 Related Work

2.1 AI Literacy for Youth and the Challenge of Contrasting AI Paradigms

AI literacy involves understanding how AI systems work, their limitations, and the human decisions shaping their design [1, 5, 19]. As adolescents increasingly encounter AI, developing clear mental models of AI behavior becomes essential [21], yet many struggle to understand how data, rules, and features shape system outcomes [1, 6, 11]. In Child–Computer Interaction and HCI, hands-on pedagogical tools—such as Teachable Machine [4], Machine Learning for Kids [20], Dataland [18], and Google AI Quests [10]—support engagement with data-driven AI through interactive and contextualized learning. While these tools effectively introduce data-driven AI, they often present AI as a single learning pipeline and rarely contrast symbolic (rule-based) and connectionist (data-driven) reasoning paradigms [9, 15]. This lack of contrast matters to, as learners struggle to reconcile deterministic rule execution with probabilistic inference [6, 11]. Only a few youth-oriented learning experiences support the construction, testing, and comparison of both paradigms within a coherent progression [5, 35].

2.2 Tangible Interaction and Constructionist Approaches to Abstract AI Concepts

When AI systems operate implicitly or present only final predictions, adolescents often struggle to reason about the abstract concepts—such as *rules*, *models*, and *data quality*—that shape system behavior. Tangible User Interfaces (TUIs) address this challenge by giving physical form to digital information, making intermediate states visible and enabling embodied manipulation of abstract concepts [8, 34]. Building on these principles, tangible programming blocks [13], unplugged computing [3], tangible games [29], and embodied AI systems [37] have supported learning in programming and AI by grounding computation in physical action and shifting focus from learning programming syntax to developing conceptual reasoning. However, much of this work emphasizes a single AI reasoning paradigm—either rule-based or data-driven—without enabling learners to directly compare them in context. Our work extends prior tangible and constructionist approaches by making both symbolic and connectionist reasoning physically inspectable. Thus, we support learners with game cards that embody important AI concepts like rules and training data so that they can use these external models for side-by-side comparison and reflection to nurture their reasoning.

2.3 Situated and Game-Based Learning for Iteration and Reflection

Games and narratives are widely used in education to foster engagement, exploration, and meaningful, goal-directed learning [17, 25, 28]. In tangible systems for AI education, narrative structures—including quest-based scaffolding [10, 36] and storytelling [32]—frame AI concepts within game mechanics that contextualize player

missions and activities. Beyond engagement, these environments support failure, iteration, and revision, which are essential for developing accurate mental models of AI. Human–AI interaction research shows that AI behavior emerges through iterative human decisions about rules, data, feedback, and evaluation [22, 23]. Learning experiences that foreground iteration and comparison help learners attribute system outcomes to these decisions rather than to arbitrariness.

Building on these insights, our work uses narrative progression to scaffold learners making a conceptual shift. Learners first experience the strengths and limits of manual rules. Then they encounter an in-game need to adopt and critically evaluate data-driven learning. We present a tangible-interaction board game, *Pokémon AI Adventure*, that addresses this gap. It combines tangible model-building, structured comparison, and narrative motivation. This helps adolescents explain AI behavior, diagnose the causes of model failure, and form reflections on human responsibility in shaping system outcomes.

3 System Design and Implementation

Our objective is to foster adolescents' AI literacy by supporting understanding of the design, training, and evaluation of rule-based and data-driven AI systems through tangible, interactive gameplay. We propose design principles that guide learning experience prototyping and make AI behavior observable, comparable, and attributable to human decisions.

Design for Contrastive Understanding of AI Paradigms. Comparison supports conceptual learning by directing attention to deep differences in strategies and outcomes [27]. In AI education, a focus on a single data-driven pipeline can obscure key distinctions between symbolic and connectionist approaches [9]. Explicitly contrasting these paradigms can help adolescents reason about model failures, human control over model behavior, and sources of uncertainty in model breakdowns.

Embody AI concepts in tangible representations. To support adolescents' transition from concrete manipulation to abstract reasoning, AI learning systems should make model components such as *features* and *training data* visible and physically manipulable [2, 8]. Constructionist learning theory suggests that learners benefit from assembling and revising these building blocks through embodied action rather than abstract definitions alone [24, 26]. Our system enables learners to physically arrange, use tangible robots to scan cards to construct rule sets or training datasets, making algorithm design a concrete and inspectable process [26].

Support engagement and reflection through narrative gameplay. Game-based learning can motivate adolescents to actively explore AI concepts and persist through challenges when learning is grounded in meaningful learning objectives and narrative progression [10, 25]. Accordingly, AI learning experiences can embed modeling tasks within game-based storylines that frame errors as informative and encourage iterative experimentation [32].

3.1 Game Design

We designed *Pokémon AI Adventure* to support adolescents' AI literacy. Using tangible, narrative-driven gameplay, we enable hands-on comparison of rule-based and data-driven AI to make abstract

concepts such as rules, features, and training data concrete and inspectable.

3.1.1 Game Background. Our game uses the Pokémon universe¹ as a motivating narrative framework for AI learning. Set in the fictional world, Novara: a newly discovered continent with diverse ecological regions. In Novara, environmental changes have made Pokémon behavior difficult to predict, and so, prompt Professor Oak to launch an AI-driven research expedition. Players explore Pokémon behavior through region-based challenges using a tangible, robotic toy to assist them. They are introduced to two AI approaches through two toy robots: *RuleBot*, which follows explicit human-authored rules, and *DataBot*, which learns patterns from examples. Using physical cards as tokens that can be scanned by the handheld robots, players configure RuleBot and DataBot to identify Pokémon across regions. Pokémon are characterized by observable features, environmental cues, and patterns in training data, embedded within narrative missions that encourage experimentation. Through successes and unexpected outcomes, learners connect tangible interaction, narrative context, and iterative feedback to understand how different AI designs shape behavior.

3.1.2 Game Flow. The game follows a structured progression designed to scaffold adolescents' understanding of AI through repeated cycles of *design, testing, observation, and reflection*. Such iterative learning loops support conceptual change by helping learners construct and revise mental models through experience and feedback. Gameplay begins with an introduction and continues through four ecological regions, each building on prior player actions while introducing new AI concepts in line with constructivist and experiential learning approaches [12, 30].

- **Orientation.** Players begin by exploring a physical guidebook that introduces the world of Novara, core Pokémon types, and the overall research mission. This orientation phase ensures accessibility for participants who may be unfamiliar with the Pokémon universe, while establishing a shared reference frame for features, categories, and classification tasks that will be revisited throughout the game.
- **Zone 1: Clearview Meadow.** Players encounter an environmental crisis in which wildfires are spreading across the grasslands. To prevent further damage, players must teach *RuleBot* to correctly identify Fire-type Pokémon. They construct a "Fire Pokémon Plan" by scanning *Clue Cards* that define explicit, human-authored rules. Players iteratively test their rules in the Lab Zone, observing how overly strict or overly general conditions lead to misclassification, before deploying the model in the Play Zone to identify wild Pokémon. This zone introduces symbolic AI reasoning and highlights the direct relationship between rule design and system behavior [31].
- **Zone 2: Azure Coast.** A sea storm forces multiple Pokémon types to gather in the same area, even though different types cannot safely coexist. Players extend their rule-based approach to distinguish among four Pokémon types by refining and ordering rules to resolve classification conflicts. This escalation foregrounds the limitations of handcrafted logic, highlighting

¹See <https://www.pokemon.com/us/pokedex>, last accessed February 2026.

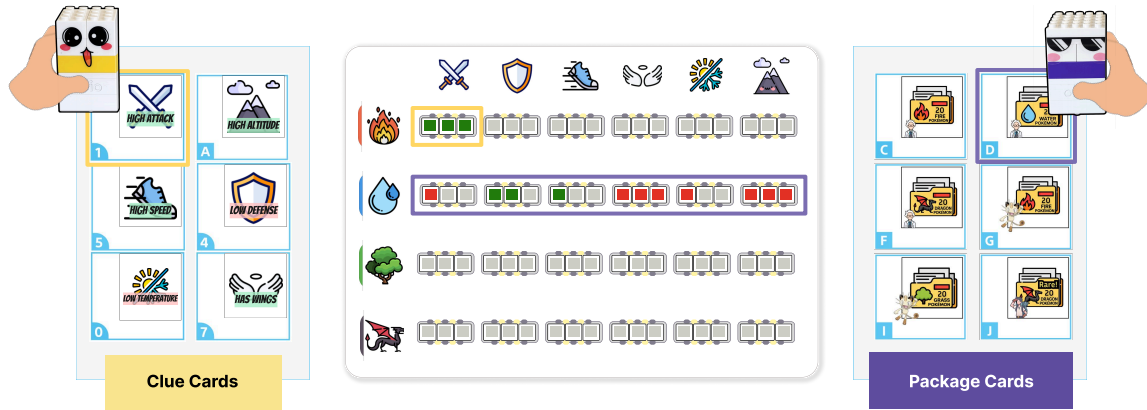


Figure 2: Building rule-based (left) and data-driven (right) AI with Clue Cards representing individual feature selections and Package Cards representing grouped training examples, reflected in the LED visualization of the model state.

trade-offs between coverage, precision, and interpretability as rule systems scale to more complex classification tasks [19].

- Zone 3: Whispering Woods.** Players change which robot they use, to *DataBot*, as the narrative introduces data-driven learning. The story reveals that the character, Meowth, has corrupted several training packages, creating uncertainty about the integrity of the available data. Players examine paired *Package Cards* (A/B) for each Pokémon type, inspect their contents, and reason about which dataset is cleaner and more representative. This zone introduces the concepts of *training data* and *data quality*, encouraging learners to identify noise or inconsistency and anticipate how data composition may influence model behavior prior to training.
- Zone 4: Sunrise Desert.** To safely cross the desert, players face a resource constraint: they may train *DataBot* using only four *Package Cards*. From a larger pool of available packages, players curate a limited training dataset by selecting and scanning cards. This design foregrounds trade-offs in dataset selection, prompting learners to reason about coverage, balance, and bias. By experimenting with different combinations, players observe how data curation choices shape *DataBot*'s predictions, reinforcing the role of human judgment in data-driven AI systems.

3.1.3 Tangible Interaction Design. The system employs tangible interaction to concretize abstract AI concepts in game tokens. *Clue Cards* and *Package Cards* create tangible tokens for controlling model design decisions. Players tangibly control the robot to scan these so that rule specification and dataset selection are embodied in explicit player choice and a visible process. As shown in Figure 2, the LED Light Board functions as a tangible representation of the AI's internal state. During early gameplay, when players construct Fire-type rules, scanning *Clue Cards* with *RuleBot* activates corresponding feature slots on the board, with color encoding (red and green) indicating low and high feature values. In the second lab, where players design rules for multiple Pokémon types, the visualization is extended to encode feature importance using both color and the number of illuminated LEDs (One to three lights indicate the card order), supporting reasoning about weighted decision logic.

When *DataBot* is introduced, the same visual language is preserved to maintain conceptual continuity; however, scanning a single *Package Card* reveals an entire learned pattern derived from training data rather than manually specified rules. By enabling learners to manipulate physical artifacts and receive immediate, responsive system feedback, the design supports constructionist learning practices that facilitate explanation, debugging, and iterative model refinement.

3.2 Technical Implementation

The system integrates tangible hardware with a digital game engine to support tangible AI interaction. A Sony *toio* robot² functions as the physical AI agent, navigating the game board, scanning cards, and expressing outcomes through motion and sound (e.g., success dances or failure shakes). We use *toio* Simple Cards and Simple Maps augmented with custom-printed symbols to represent Clue Cards, Package Cards, and Pokémon. The robot's built-in sensors read standard card IDs, while map position IDs enable localization and movement across the game space. A model visualization board driven by an ESP32 and WS2812B addressable RGB LEDs reflects real-time model state changes. The digital game interface is implemented in Unity (version 6000.3.0b4), which coordinates robot behavior, LED feedback, and on-screen visualizations of predictions, outcomes, and narrative events.

4 Demo Experience at CHI

We will present our tangible, interactive Pokémon AI Adventure game, which attendees can play hands-on. While designed primarily for adolescents, the game is open to players of all ages.

Attendee Experience. Participants begin at the *Orientation* level, where they are introduced to the game narrative, components, and how to use the robots to scan *Clue Cards* and *Package Cards*. The experience then progresses through four levels (*Zone 1* to *Zone 4*), during which participants complete the corresponding tasks described in Sec. 3.1.2. In each level, participants train their robots to

²See <https://toio.io/>, last accessed February 2026.

recognize different Pokémon types by scanning rule-based or data-driven cards while navigating the game board. The LED light board provides immediate visual feedback using color and the number of illuminated LEDs. At the end of each round, participants test their robot's learned rules on a randomly encountered "wild" Pokémon.

Demo Setup. The demo will be presented at a small standing booth with a table and a chair. The game board and tangible artifacts will be arranged on the table for hands-on interaction. Although designed for a single player, small groups may gather to discuss strategies and observe game play. Participants may join at any point and explore any zone at their own pace, with each zone's interactions typically lasting under five minutes. A digital version of the game will be available on an iPad when the tangible setup is occupied.

Safety and Comfort. The demo uses lightweight robots and tangible artifacts in a controlled tabletop setting, posing minimal safety risk. Facilitators will supervise interactions and provide guidance, and participants may stop the experience at any time.

5 Conclusion

We present *Pokémon AI Adventure*, a tangible interactive game that supports adolescents' AI literacy through hands-on construction and comparison of rule-based and data-driven AI systems. The game combines physical model building with real-time robotic feedback, enabling learners to explore how rules and data shape AI behavior. At CHI, attendees will experience Pokémon AI Adventure by constructing models with physical cards, testing them in real-time capture tasks, and reflecting on how human design decisions affect AI behavior and performance.

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